Lesson Notes

MODULE 4 | LESSON 1

DERIVATIVES, WITH AN EMPHASIS ON OPTIONS

Reading Time	120 minutes
Prior Knowledge	Bonds, Stocks, Linearity
Keywords	Derivative, Obligation, Futures, Forwards, Options, Vanilla, Exotic, Call, Put, Premium, Payoff

In the previous modules, we discussed various spot instruments: bonds, stocks, and ETFs. In this module, we switch to covering derivatives, specifically options. The option's value derives from its underlying. As a derivative, options provide choice for investors by locking in a certain price for a future transaction. In this lesson, we'll cover the basics of option types and payoffs.

1. Derivatives as a Whole

Up until now, all the securities we've seen are known as **spot securities**. We turn now to a new type of security, which is called a derivative. A **derivative** is a security whose value depends on the value of another security. It's a derivative because its value is *derived* from the prices of another security. If we know the value of the underlying security, then through the mathematics of stochastic processes and the art of securities hedging and replication, we are likely going to know the price of the derivative. More precisely, a derivative is a security whose value depends on another security or even another measurable quantity. The value of a derivative is known precisely at a given point in time, either at the creation of the derivative or at its expiration.

There are essentially four types of derivatives:

- Futures
- Forwards
- Swaps
- Options

We will save swaps for another course. In this module, we will introduce futures, forwards, and options.

Before we jump into derivatives, note that all derivatives have an expiration date. This is different from fixed income products like bonds, which have a maturity date, and from equities or ETFs, which have no end date; equities do not have a date where they pay back any principal. However, all derivatives have an expiration date after which the derivative expires, and either it 1.) is converted to a security or cash flow equivalent, or 2.) expires worthless! In the next section, we'll begin our look at derivatives with futures. Be sure to read Chapter 12 of Saylor.

2. Futures & Forwards

When you buy stock, you are agreeing to a price and a settlement that occurs now (in reality, the settlement may be a few business days after the trade date). In the U.S., beginning on May 28, 2024, the settlement date for purchasing stocks is $T+1$, down from $T+2$.

A futures contract is an agreement between two parties to buy or sell an asset at a certain time in the future for a certain price. Futures contracts trade on an exchange. One of the largest exchanges on which futures contracts are traded is the CME Group, which is the result of the merger between the Chicago Board of Trade (CBOT) and the Chicago Mercantile Exchange (CME). Other large exchanges for trading futures are Eurex, InterContinental Exchange, Tokyo Financial Exchange, and more.

Figure 1: The Chicago Board of Trade Building in Chicago.

Futures contracts deal with a variety of underlying assets: gold, tin, copper, sugar, pork belly, live cattle, stock indices, currencies, Treasury bonds, interest rates, and more.

Futures contracts are standardized in the sense that the exchange must specify the elements that characterize the exact nature of the agreement between the two parties involved. These normally consists of the following:

- Characteristic of the assets. For some commodities, there may exist several grades of the same product, and the exchange decides which grades of the commodity can be accepted. For other underlying assets like financial contracts, things are definitely less ambiguous.
- Contract size. In this case the exchange specifies the amount of assets that must be delivered to fulfill a contract. This is actually a very important decision because if the minimum size is too large for many investors, then it would be too expensive for them. On the other hand, if the contract's size is too small, investors may be forced to transact on many contracts with higher associated costs.
- Delivery arrangements. The contract usually indicates where delivery will be made. You can imagine that this is very important for commodities that are perishable and also when considering transportation costs.
- Delivery months. The exchange usually specifies the months and the exact period during the month when delivery can be made, which can vary among contracts.
- Price quotes. The exchange decides how prices will be quoted. In addition, for many contracts, the exchange could also indicate daily price movement limits. Price limits are set to prevent large price movements, particularly in the case of speculative investments.

From the description we offered, the student should have recognized that a futures contract is a derivative because it has an underlying asset.

When we buy a stock, the position is settled within a given period. This period, as of May 28, 2024, in the U.S., is $T + 1$, i.e., the settlement happens one business day after the day on which the trade took place. However, we might buy a future today, but the settlement/delivery could be in 3 months, 6 months, or even further into the future.

We must stress that a futures contract **obligates** the buyer of the future to proceed with the purchase, and it **obligates** the seller of the future to proceed with the sale. This derivative is a legally binding agreement between both parties. Each side agrees to honor the terms, even though the prices may move in such a way that the terms become very unfavorable to one party.

Example. *Consider the futures contract from the buyer's perspective. Suppose the buyer holds a futures contract to buy 1,000 kilograms of aluminum at 3,200 euros. The delivery is for three months' time. Over the course of those three months, demand decreases, lowering the price to 2,500 euros. The buyer would prefer if they did not agree to the higher price since he could buy 1,000 kg in the spot market for 2,500 euro instead of 3,200 euro. Nevertheless, the current market price is irrelevant in deciding whether the trade should still take place. Futures are obligations. Counterparties are not allowed to back out of the futures contract*. Likewise, suppose there were a supply disruption instead, hiking the price to 4,000 euros. Now the buyer is happy to pay 3,200 euros because everyone in the current market is paying 25% higher. The seller, on the other hand, would prefer to sell at the higher price. Again, each side must honor the obligation*.

As a matter of fact, only a small amount of money changes hands until the future is actually traded. These upfront costs are related to initial margins, exchange fees, and requests for variation margins, i.e., additional funds to keep the margin at the current initial margin level.

Let's look at an example that illustrates these aspects.

Example. *It is now June 30, 20XX. Suppose trader A wants to buy three December gold futures contracts. The current price of* *gold is \$2,370 per ounce. The future establishes a size of 100 ounces. Then, trader A has committed to buy a total of 300 ounces of gold at a price of \$2,370 per ounce. This transaction takes place via a broker. The value of the transaction is \$711,000. The broker will ask trader A to deposit a certain amount of money in a margin account, and this amount must be deposited at the time the contract is entered, i.e., June 30, 20XX. This margin is known as the initial margin. We assume that the margin required is \$300,000. From this moment on, the margin account is adjusted each day to reflect trader A's gains or losses (daily settlement). On July 1, 20XX, the futures price has dropped from \$2,370 per ounce to \$2,350 per ounce. Trader A has lost \$300 X 20 = \$6000 (note that the 300 ounces that he contracted to buy for \$2370 per ounce can now be sold for \$2,350 per once). The balance in the margin account goes from \$300,000 to \$294,000. If on July 2, 20XX, the price of gold goes up to \$2,380 per ounce, the balance in the margin account increases by \$$300 \times 30 =$ 9,000 and becomes \$303,000. Trader A is entitled to withdraw any balance in excess of the initial margin. Also, there is a maintenance margin. If the balance in the margin account falls below the maintenance margin, trader A receives a call asking him to deposit additional funds (usually by the end of the next day) to re-establish the initial margin account. These additional margins are called variation margins. If trader A fails to do so, the broker will close out trader A's position.*

Why does the exchange require this margin? The reason is an old friend we've discussed: credit risk. Without the margin, a buyer could theoretically show up on the delivery date with insufficient funds to proceed with the trade. Consistent margin calls by the exchange ensure this doesn't happen. In doing so, exchanges help to mitigate credit risk.

Forwards are similar to futures contracts except that forward contracts are not exchange-traded. Forwards trade on over-the-counter markets. They may have custom dates and notional amounts since they are legal agreements between two counterparties. Compare this to futures, which are exchange traded. In a futures contract, the counterparty is actually the exchange itself. This allows the exchange to measure and monitor the credit of each side, requiring margin calls where needed and eliminating credit risk. Therefore, forwards have counterparty risk, whereas futures do not. To repeat: exchanges require margin calls to provide ready cash so that one side of the futures will have sufficient cash on hand to carry out the trade. This distinction is why forwards have more credit risk than the corresponding futures.

Example. *Company PQM is based in the U.S. and sells a large percentage of their products in Europe. Company PQM is paid in euros and expects to receive a payment of 40 million euros in 4 months. Currently, the exchange rate between the EUR and the USD is 1.08 USD per 1 EUR. Company PQM is afraid that the EUR may depreciate vs. the USD. Think of this: currently, PQM stands to receive* $40,000,000 \times 1.08 = 43,200,000.00$ *USD in 4 months. If the EUR depreciates toward the USD--say that, for instance, the exchange rate becomes 1 USD per 1 EUR--then company PQM would receive only $40,000,000. To avoid this situation, company PQM enters in a forward contract with a counterparty to sell $40,000,000 EUR at a rate of 1.077 USD per 1 EUR. This is an example of using a forward contract to hedge an exposure in a foreign currency. Of course, in 4 months, the EUR could appreciate vs. the USD (say the exchange rate goes from 1.08 USD per 1 EUR to 1.10 USD per 1 EUR) and the company would lose this upside, but the company is most concerned about the possibility of the EUR depreciating toward the dollar and so hedges its exposure using a forward contract.*

3. Options

An option is a derivative that provides the holder with optionality, or choice, in whether to trade a security in the future at a specific price. The two basic types of options are calls and puts.

A **call option** grants the holder, for a limited period of time, the right, but not the obligation, to purchase the underlying asset at a fixed price. That is, the holder calls in the underlying at the price that was fixed.

A **put option** grants the holder, for a limited period of time, the right, but not the responsibility, to sell the underlying asset at a fixed price. That is, the holder puts the security on sale at a fixed price.

The fixed price common to calls and puts is known as the **strike price**, or **strike level**, or simply the **strike**. It is usually denoted by K.

Where does the option buyer purchase the option? This would most likely be on an options exchange. Vanilla options trade on exchanges. A **vanilla** option is the name given to an option that has a standard payoff. An **exotic** option is the name given to an option that has a payoff different from the standard one.

Let us consider the payoff of the classic type of option, calls and puts.

An option costs money to purchase. The price of any option is known as the **premium**. The premium is the price the buyer pays to enter a call or to buy a put. But note that the price of a call and a put with the same time to expiration, strike, and underlying assets are not the same in general.

Once an option buyer pays the premium, they own the option. They will also never lose more money than the premium. Unlike stocks, options cannot be bought using margin. The premium must be paid in its entirety. Does that mean that options do not have leverage? Absolutely not. Options have lots of leverage--something we'll discuss later in this module.

For now, let's suppose that the option is of a European style, i.e., it can only be exercised at its expiration. The payoff of a call option is equal to the maximum of the final stock price minus the strike level, or zero. In formula

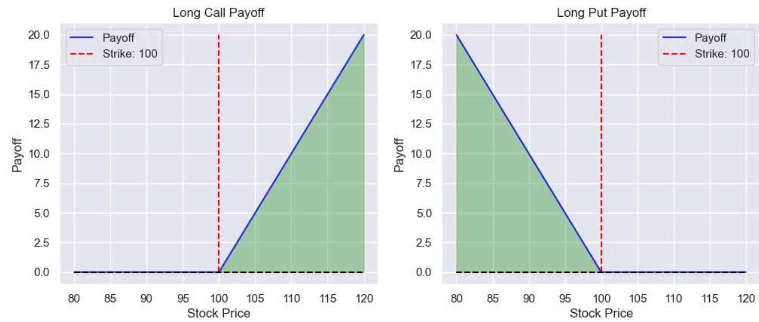
$$\textbf{call payoff} = \max\{S_T - K, 0\},$$

where T is the time at expiration, S_T is the price of the underlying at time T, and K is the strike price.

The payoff of a European style put option is equal to the maximum of the strike level minus the final stock price, or zero. Here is the formula:

$$\textbf{put payoff} = \max\{K - S_T, 0\}.$$

Figure 2: Long Call and Long Put Payoffs



You should have noticed from the formulae that the payoff of an option is known only at expiration. And you should also have paid attention to the fact that payoffs have a choice because the $\max\{\cdot\}$ function is applied and therefore calls and puts have non-negative payoffs. Be sure not to confuse this fact with the value of an option that, like for many other securities, can fluctuate over time.

Let's consider a European call option. Buying an option means you are long the option. It also means you are the option holder. For the long option side, you paid a premium to enter the position, and your payoff is given by the above formula. The P&L for a call option buyer is

$$\textbf{call PnL} = \max\{S_T – K, 0\} – \textbf{premium}.$$

While the payoff term is never negative, the P&L can be negative because of the premium. Indeed, even if the payoff term is positive, if it is smaller than the premium, the P&L can still be negative. For a European put, we instead have

$$\textbf{put PnL} = \max\{K - S_T, 0\} – \textbf{premium},$$

and the same remarks about the fact that the P&L could be negative apply in the case of a European call.

We've been discussing options from the buyer's perspective. The **option writer** is selling the option and is considered to be **short** the option. Let's look at the payoff and P&L formulae from their perspective. The payoffs for long and short calls, respectively, are:

$$\text{long call payoff} = \max\{S_T - K, 0\}$$

and

$$\text{short call payoff} = -\max\{S_T - K, 0\}.$$

Note that the sum of the two payoffs is 0. **Options are a zero-sum game**. The gains by one side equal the losses to the other side. Options are contracts between two counterparties, and the company or third parties are not involved in the writing or P&L of these options.

Likewise, the formulae for the payoff of long and short puts, respectively, are:

$$\text{long put payoff } = \max\{K - S_T, 0\},$$

and

$$\text{short put payoff } = -\max\{K - S_T, 0\}$$

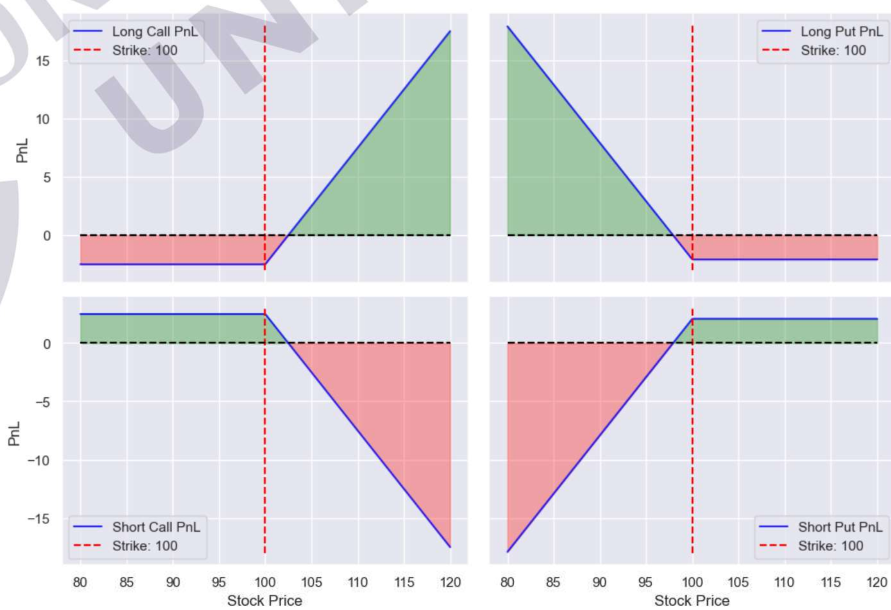
Let's now consider the P&L. Here are the formulae for the P&L of calls:

$$\text{long call PnL } = \max\{S_T - K, 0\} - \text{premium}$$

and

$$\text{short call PnL } = -\max(S_T - K, 0\} + \text{premium}.$$

Figure 3: Long/Short Call/Put PnLs - The shape of the plots resemble a diamond.



Just as we saw for payoffs, the P&L for options is a zero-sum game. The gains to one side are losses to the other side. This is true not only for the P&L but for each of its components: payoffs and premiums.

An option's **intrinsic value** is its payoff at any given time. There are three categories of intrinsic value:

- **in the money** (ITM) when the payoff is positive.
- **at the money** (ATM) when the payoff is 0 because the strike and stock price are equal.
- **out of the money** (OTM) when the payoff is 0 because we are on the flat part of the payoff diagram.

For a call, $S > K$ makes an option ITM. For a put, $K > S$ makes an option ITM.

For a call, $S < K$ makes an option OTM. For a put, $S > K$ makes an option OTM.

For the call writer, the premium is income rather than a cost. The premium is the most that the writer can actually make. If the option expires ITM, then the writer will make less than the premium amount. Indeed, it's possible for the writer to have a huge loss if the stock price greatly increases. For the call buyer, the premium is a fixed cost. If the option expires OTM, then the holder will make nothing. Indeed, it is easy to lose your entire premium by purchasing an option that expires out of the money.

Please be sure to distinguish an options premium from its payoff. The premium is the price that you would pay for the option for the weeks or months before its expiration date. The payoff of the option is only applicable at the option's expiration date.

The premium itself can be understood as having two sources of value:

- Intrinsic value: this is the same as the payoff if the option could be immediately exercised.
- Time value: this is the time value of the option.

Hence:

$$\textbf{Premium} = \textbf{Intrinsic Value} + \textbf{Time Value}$$

The premium should always be positive. If premiums were 0 (or negative), then you would basically have a free option. That means there would certainly be an arbitrage. You would get something today that costs nothing and a chance to get a positive payoff in the future. This is effectively a free lottery ticket. In order for the left-hand side to be positive, the right-hand side must also always be positive. That means the sum of the intrinsic value and time value must be positive. Imagine you had an option that is deeply OTM. Clearly, there would be no intrinsic value. All of the cost in the premium would come from time value.

Next, we analyze the factors that affect the prices of options.

4. Option Dependency 1: Underlying Price

The options' underlying security is the security from which the option derives its value. Let's suppose we have stock options. Then, the specific stock, say Netflix (NFLX), is an explicit part of the option's terms. During the life of the option, we would use the dynamically changing stock price in a model that helps us price the option's premium. At the option's expiration, we use the stock's closing price, which we compare to the strike to determine if the option is in the money. For calls, as the price of the underlying increases, the value of the call increases. Why? Looking at the call's payoff, we have the term $S_T - K$. A higher stock price will lead to a higher payoff.

For puts, as the price of the underlying increases, the value of the put decreases. Why? Looking at the put's payoff, we have the term $K - S_T$. A higher stock price will lead to a lower payoff.

5. Option Dependency 2: Strike Price

Unlike the stock price, the strike does not change. The strike represents a hurdle for the stock price that must be on a particular side for the option to be in the money. Notice that for both calls and puts, the signs of the stock price and strike are always opposite. Therefore, the higher the strike, the less a call is worth; the lower the strike, the more a call is worth.

The call's payoff $S_T - K$ will be maximized when we choose the lowest strike.

The put's payoff $K - S_T$ will be maximized when we choose the highest strike.

The strike level is something the option buyer selects at the time of purchasing the option and must live with while the option position is held.

Suppose you bought two different call options: one with a strike of 110 and another with a strike of 120. Suppose the final stock price is 118. The first option is 118-110 = $8 in the money. The second strike is out of the money because 118 < 120. Choose your strikes carefully!

For a call option, we can see that the higher the strike price, the harder it is to produce a payoff. Clearly then, these options will be less expensive because they are less likely to be in the money than options (on the same underlying) with lower strikes.

Note that the hurdle analogy is not a one-time phenomenon. If the stock price in our previous examples dropped down to 105, then neither the 110-strike nor the 120-stirke option would be in the money. Be sure you understand the payoff formula: the stock price used is at expiration, given at time T. This is our next dependency.

6. Option Dependency 3: Expiration Time

The stock price that is used to determine the payoff is the stock at time T, the time at which the option expires. The option's expiration is the end of the story. It is very objective: either the option is in the money, or it does not pay off. That is, if an option ends up at the money or out of the money, the option holder has a security that is effectively worthless. How could it lose all its value? The option grants a powerful right: the right to trade at a level that is favorable to the market. Unlike a future, it does not force the holder to execute at this price. If this were to continue indefinitely, options would be difficult to price and likely too expensive. Who knows what can happen many years from now? The option expires at a certain time, which helps to contain the uncertainty by shortening the interval of time.

Of all the random variables involved, time is the easiest to understand. It only moves in one direction and at a constant rate. As options draw closer to their expiration date, the tendency is that they lose time value. All else being equal, their overall cost is likely declining. Note: There are exceptions that will be addressed in a future course, but for now, think of this as the general trend.

If you were to compare two options that are near the money and that are similar in all aspects EXCEPT that one has a longer expiration date, then the longer-dated option will be worth more, simply because it offers more optionality. Note that this relationship is the same whether the option is a call or a put.

Let's reiterate: the optionality in options is a limited-time offer. That means that at some future point in time, the optionality of the holder to trade at the strike expires. This means that it's possible to hold an option to a point where it expired worthless. When the stock price exceeds the strike price, the calls are in the money, but the puts are out of the money.

If the option expires in the money, the holder will undoubtedly exercise the option. If the option is out of the money, then the payoff is 0 and the option is not exercised. It is in this last case that 100% loss of investment is possible. When it expires in the money, any gain is possible.

The price at which you trade could be quite favorable to the current price and if the option is deep in the money. Indeed, for call options, the stock price is unbounded, so there are infinite stock price paths that exceed the strike level.

Typically, the more time you have until expiration, the more value the option has. The less time you have, the less valuable it becomes. Time value affects values when there is a specified expiration date or deadline. Imagine that you are selling a concert ticket to see a very popular band. Imagine that you have one week before the concert to sell it. You may find that you can charge a much higher price. Now imagine the concert starts in 5 minutes. Once the concert starts, the ticket will become worthless; the time value of this ticket has decreased. It is the prospective ticket buyer who benefits from this, as the seller becomes very highly motivated to sell the ticket. Otherwise, they could lose the cost of the ticket.

Likewise, as you hold an option, the closer you get to the expiration, all else being equal, the option is worth less. This simply means that the greater the amount of time, the greater the possibility that you could reach higher levels of being in the money.

7. Option Dependencies 4 & 5: Risk-Free Rate and Dividend Yield

Options depend on the interest rate that applies to the period that matches the length of time of the option's expiration. We refer to the rates as r. Similarly, options depend on the income-generating power of the underlying stock, namely the dividend yield. We refer to the dividend yield as q. We will postpone the details of option prices dependent on interest rates and dividend yields. The reason is that to understand this, we will actually want to know how an option is hedged, and that comes later in the program. For now, let's just state the relationships:

- Calls increase in value if interest rates increase OR if dividend yields decrease.
- Puts decrease in value if interest rates increase OR if dividend yields decrease.

Some students struggle to understand these relationships. For instance, why does the value of a call option goes up when the risk-free rate goes up? While options will be dealt more thoroughly in later courses, let's look in this problem as it is simple and it teaches us how to think about these issues.

Say that we want to buy 100 shares of stock FGH and FGH shares currently trade at $100 per share. We plan to stay invested for 1 year as we expect that by then FGH shares will have gone high enough for us to sell them and realize a profit. A 1 year American style call option on FGH can currently be purchased for $11, instead.

We have two ways to purchase those 100 shares:

(1) We buy the 100 shares which would cost us $10,000;

(2) We buy 1 call contract which would cost us $100 X $11 = $1,100. And we could deposit the difference, $8,900, in a saving account where the money earns an interest rate. If the interest rate in our saving account were, say 4%, then we could earn $8,900 X 4% = $356 a year.

Which strategy is more appealing? Note that the upside potential is the same. So, the second strategy looks more compelling and hence call options are desirable in this respect. Furthermore they become more advantageous the higher the risk-free interest rate is. Hence, if call options are more desirable, it means that more investors will want them, and if more investors want them, the bidding price will go up and so the value of the call option rises.

8. Option Dependency 6: Volatility

We have one remaining option: dependency. In fact, this is the most important influence on an option’s price. It is also the most complex. It is the volatility of the underlying. In other words, the option does not merely depend on S, K, T, r, and q. Typically, K is fixed, and the other value can easily be read by market prices or data. Why is the underlying so important? Because the nature of the underlying influences what the call option is worth.

The option's value is very dependent upon the specific underlying stock. What is so important about the stock? Certainly, we can see that the price of a stock is important because the price at the option's expiration is in the formula itself for the payoff. What does not appear in the formula, however, but is the most important input into an option, is the stock's volatility. If there is more volatility, there is a chance for the stock to have more extreme values. Will this make the option's values more extreme? It may appear so because volatility is agnostic about direction: more volatility means more movement in either direction. So will higher volatility cause the option to be worth more? The answer is a resounding yes. We’ll dive into the reasons why in the next lesson.

9. How are options exercised?

Arguably, most students are aware of how to buy and sell shares. But how about options? Say that you own a European style call option to buy 100 shares of MNQ for $60 a share. The stock has risen to $68 per share and the option is expiring today. Clearly, you want to exercise the call as *it is in the money*. How do you do that?

- First of all, we must be aware that the OCC, the **Options Clearing Corporation**, has the control over all exercises and assignments after the trades are made. The term assignment will become clear after you read below.
- The first step you must take is to instruct your broker to exercise the option for you. Typically, you need to have a broker that supports options trading, whose fees are adequate for your needs and provides a suitable platform usability together with customer service, and educational resources. Keep in mind that to trade options, the broker

must approve you and the broker may have several levels of approvals where the riskiest strategies could be available to selected sophisticated investors only. However, here, we are just dealing with basic option contracts.

- At this point, the broker will notify his administrative staff to exercise and buy the stock. The administrative staff knows all the steps required to exercise options in a manner conformal to the rules.
- Hence, the order is then sent to the OCC: exercise one contract of the MNQ June 60 call series.
- In this step, the OCC selects a firm who is short the MNQ June 60 call (this is what is known as the **assignment process**). The assignment could be a simple random assignment or follow a *first in, first out* basis.
- The firm that was selected in the assignment process is required to deliver 100 shares of MNQ at $60 per share to the firm that exercised the option, i.e., you!
- The OCC is usually indifferent about the method of delivery as long as the 100 shares of MNQ are delivered and paid $60 per share. You, as the person who exercised the call can keep the stock in your account if you want to, but must pay cash or margin it fully. Clearly you may want to sell it in the open market for a price higher than $60 and realize a profit. If you decide to hold on to the shares, then you are exposed to market risk as the shares, after climbing could even go down in price afterwards and if they went below $60 you would realize a loss if you sold your shares.
- However, if you have a margin account with your broker, you can sell the shares immediately.

The **physical settlement** of options contracts is the most common form of settlement and, as the name suggests, it involves the physical or actual delivery of the underlying security at settlement. However, one should be aware that there is also such thing as a **cash settlement**. This form of settlement occurs when cash exchanges hands at settlement instead of an underlying security or physical commodity. In our example, you would receive $8 per share or $$800 for the full contract minus commissions and fees. In the case of commodities there could be an economic reason for a cash settlement since, in doing so, we avoid the costs of storage. Cash settlement is also the primary settlement in the case of index options for the simple reason that an index is not deliverable.

10. Conclusion

In this lesson, we addressed the types and payoffs of vanilla options. In the next lesson, we'll examine payoffs more closely by looking at two key features: leverage and nonlinearity.